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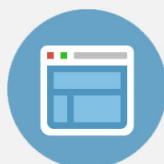
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Optimization of quantum mechanical molecular mechanical partitioning schemes: Gaussian delocalization of molecular mechanical charges and the double link atom method

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Two new techniques for modeling chemical processes in condensed phases with combined quantum mechanical and molecular mechanical (QM/MM) potentials are introduced and tested on small, model compounds. The first technique, the double link atom (DLA) method, is an extension of the traditional, single link atom (SLA) method to avoid some of the problems with the latter method. These problems are primarily electrostatic, as the SLA method can produce an unphysical overall charge or dipole. The second technique, the delocalized Gaussian MM charge (DGMM) method, is an empirical way to include the delocalized character of the electron density of atoms in the MM region. This can be important for the electrostatic interaction of the QM region with nearby atoms in the MM region, and it can simplify the rules governing which classical interactions are included in the energies and forces. Even for very short distances, the DGMM method does not require the neglect of the MM host in the QM calculation. The DGMM method can be used for modeling reactions in solution, and it can be combined with methods such as the link atom, frozen orbital, or pseudopotential methods for terminating the QM region at a covalent bond. The DLA and the DGMM methods have been combined effectively. Presented here are tests on small, model systems that mimic properties important for reactions in proteins, in particular rotational barriers, proton affinities, and deprotonation energies. The new methods yield improved energetics for model compounds, *vis-à-vis* a point-MM-charge and SLA treatment. © 2002 American Institute of Physics. [DOI: 10.1063/1.1520134]

I. INTRODUCTION

Ab initio quantum mechanical (QM) calculations provide accurate information regarding the structure and reactivity of molecules. Ground and excited state energies can be calculated; geometry optimization to stable species and transition states can be performed. In spite of better and faster algorithms,^{1–3} and the tremendous increase in computational resources over the years, QM methods are limited by the size of the system. A maximum of about 2000 basis functions can be handled at the present time. Molecular mechanics (MM) methods, on the other hand, provide structure and dynamics of thousands of atoms at a fraction of the cost required to carry out QM calculations.⁴ Generation of MM force fields requires extensive parametrization, but currently they are not suitable for examining bond making and bond breaking processes since they have the wrong analytic form away from the equilibrium structure. By combining QM with MM in a

suitable manner, one can study a host of systems intractable by QM, or not suitable to be studied by MM. The basic principle is simple: use a QM treatment in the region of chemical reaction, use an MM treatment for the vast majority of the system, and interface the two regions. Combined QM/MM approaches have been used in studies of reactions in solution and enzyme environments.^{5–9} One of the primary difficulties for QM/MM methods is devising a realistic interface between the QM and MM regions. Two matters in particular require attention: the termination of the QM and MM fragments when the boundary between the QM and MM regions severs a covalent bond, and the treatment of electrostatic interactions between the QM and MM regions. The methods presented here address these matters.

If covalent bonds do not have to be broken in the partition, study of the system is straightforward. One can calculate the energy of the QM region using any of the standard QM techniques, and calculate the energy of the MM region using any force field. It is straightforward to compute the Coulomb and van der Waals interaction energy between QM and MM regions. The total energy of the system is the sum of these energies. However, when the boundary between the

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QM and MM regions crosses a covalent bond, as is inevitable for reactions in enzymes, artificially fragmented molecules are created and must be realistically terminated and joined. Not capping the electron density along the broken bond can produce unphysical results.¹⁰ QM/MM schemes differ in the way the electron density along the broken bond is treated. The link atom method, one of the earliest, involves saturating the QM valency with an extra atom.^{11–13} The link atom is usually a hydrogen, but other atoms or groups of atoms can be used. In fact, artificial halogen-like atoms called pseudohalogens have been used to better mimic properties such as electron donation and withdrawal.¹⁴ Methods employing frozen orbitals and pseudopotentials avoid the need to insert an extra atom, and are receiving increasing attention.^{15–18} The QM region is treated by either semiempirical (MNDO or AMI) or *ab initio* (HF and DFT) levels.^{19–23} The MM region is treated by force fields such as CHARMM²⁴ or AMBER.²⁵ Sherwood gives an excellent review of different QM/MM partitioning schemes including the issues involved in consideration of suitable force fields.²⁶ He also discusses how different QM and MM software can be combined.

Realistically joining the QM and MM regions using link atoms requires that some but not all MM energy terms involving atoms in both regions be kept, and the electrostatic interaction between the MM and QM fragments be treated appropriately. A variety of approaches for treating link atoms has been devised. Different implementations of the link atom approach have used slightly different rules, with varying degrees of success in achieving a balanced set of interactions. Singh and Kollman,¹¹ and Field, Bash, and Karplus¹² include QM/MM bonding terms when at least one atom belongs to the MM region and one belongs to the QM region. Field *et al.*¹² do not assign any charge or van der Waals parameters to the link atoms, and they do not contribute to the MM internal energy terms. Morokuma uses hydrogens as link atoms and partitions his systems into QM and MM systems (IMOMM²⁷ method), or more than one QM systems treated at different quantum levels (IMOMO²⁸ and ONIOM²⁹ methods). In IMOMM,²⁷ bond interactions between QM atoms and MM boundary atoms are not calculated. QM/MM angle and torsion interactions involving a QM atom and a MM boundary atom are calculated only when a “pure” MM atom from the MM core is also involved. The MM boundary atom coordinate set is treated as a function of the QM boundary atom coordinate set and the link atom coordinate set. Note that in this paper, we use the term “boundary atom” in a general sense, referring to any atom in the interface region.

The frozen orbital methods also vary in their implementation. In the method pioneered by Rivail *et al.*, a strictly localized orbital, obtained from calculations on related small molecules, terminates the QM fragment and is used in the QM calculations.^{30–32} Gao has devised a generalized hybrid orbital (GHO) approach.¹⁸ In this approach, hybrid orbitals on boundary atoms are retained, and some are frozen and some optimized in the QM portion of the QM/MM calculation. Reuter and co-workers have compared link atom schemes with the local self-consistent field (LSCF) method.³³ LSCF is a frozen orbital method. They conclude

that with proper care in partitioning and treatment of electrostatics, link atom and LSCF frozen orbital methods can give comparable results and neither one is systematically better than the other. Friesner and co-workers have generated a database of frozen orbitals by performing calculations on model compounds.^{17,34} The transferred orbitals are rotated and reorthonormalized. Roothans equations are modified to make sure the orbitals stay frozen. Parametrized correction terms are calculated at the MM level for QM/MM bond stretching, bending, and torsion terms. Pseudopotential approaches, in the spirit of the pseudohalogens mentioned above, have also received recent attention.¹⁵ An advantage of the pseudopotential approaches over the frozen orbital approach is that the need to reparametrize for each new system is avoided. Yang and co-workers have partitioned the QM and MM regions using a pseudobond between the QM and MM boundary atoms.¹⁵ A fluorine basis set with an effective core potential is used to model the MM boundary atom. The pseudobond has properties similar to the bond it has replaced. The parameters for the pseudobond are basis-set dependent and specific to the hybridization of the MM boundary atom, but independent of the force field. QM/MM bonding terms are calculated at the MM level when at least one QM atom and one MM atom are involved. Recently, Thiel and co-workers proposed the adjusted connection atom scheme¹⁶ to partition QM and MM regions, and made comparisons to their earlier link atom scheme.³⁵ The connection atom, approximately located at the MM boundary of the cleaved C–C bond, is parametrized to reproduce the structure and electronic properties of a methyl group. The parametrization is done at the QM level, so the parameters are transferable to any force field. The connection atom interacts as a specially parametrized C atom with other QM atoms and as a standard carbon atom with other MM atoms. The connection atom scheme, which employs semiempirical QM methods, is similar in spirit to the pseudobond scheme of Yang *et al.*,¹⁵ which employs HF and DFT methods for carrying out QM calculations.

Several methods for treating the electrostatic interactions between the QM and MM regions have been devised. The simplest approach is to mechanically embed the QM region within the MM region, as has been done by Morokuma and co-workers²⁷ and Thiel and co-workers.³⁵ Point charges are assigned to QM atoms and their electrostatic interaction with the MM atomic point charges are calculated classically. This approach does not allow any polarization of the QM region. On the other hand, including the MM atomic point charges in the one-electron integrals of the QM calculation allows for polarization of the QM region in response to the MM environment. Some authors refer to this as electrostatic embedding.^{16,26} There has been general agreement that the charges of some MM atoms near the QM/MM boundary should be excluded from the QM calculation. However, precisely which atoms to exclude is a matter of debate. Singh and Kollman exclude MM atoms within three bonds of the QM boundary atom from QM/MM electrostatic interaction.¹¹ Field, Bash, and Karplus used different cutoff schemes.¹² The best result was obtained when all MM atoms were interacting with the QM atoms except the link atom.

Payling *et al.*^{36–38} and others exclude only the MM boundary atom from the electrostatic field and we do likewise in our usual point charge implementation.¹³ We also investigate the effect of excluding all MM atoms belonging to the same neutral group as the MM boundary atom.¹³ To preserve electrostatic properties along the partitioned bond, Friesner and co-workers place a parametrized MM charge in the middle of the frozen bond.^{17,34} The parametrized MM charge has electrostatic interaction with other MM atoms. All MM charges including the charge in the middle of the frozen bond interact with the QM one electron integrals. Short-range QM/MM electrostatic interactions are minimized by having a MM electrostatic correction to the total energy. Besides the mechanical embedding scheme described earlier, Thiel and co-workers also treat the polarization of the QM atoms by MM atoms, and polarization of both QM and MM atoms by one another.^{16,35} In the link atom scheme, QM polarization is treated in two different ways.³⁵ One is to let all MM atoms polarize QM atoms except the link atom. The other is to neglect the electrostatic interaction of the MM boundary group with the QM atoms but let all other MM atoms polarize QM atoms including the link atom. In the connection atom scheme,¹⁶ the connection atom group has no electrostatic interaction with the QM region. During polarization of the MM region by the QM region, the MM boundary atoms (link atom scheme) or the atoms covalently connected to the connection atom are not polarized. But, the effect of MM polarization for reactions of interest seems to be small. Thiel *et al.* found mechanical embedding to be inadequate.³⁵ MM atoms within the two nearest-neighboring groups of the QM boundary atom do not polarize the QM region in the scheme of Yang *et al.*¹⁵ Hillier and co-workers³⁹ move charges from the MM boundary atom into the adjacent MM atoms to preserve electrostatic properties along the broken bond.

The treatment of van der Waals interactions varies. Morokuma calculates van der Waals interactions between MM boundary atoms (if more than one boundary atom is present) but van der Waals interactions between boundary atoms and QM atoms are not calculated in order to avoid double counting the interaction.²⁸ Many use the same van der Waals parameter set for both QM and MM atoms. Friesner, however, uses specially parametrized van der Waals parameters for QM atoms.^{17,34}

Geometry optimization using the combined QM/MM potential surface has also been dealt with. Kollman and co-workers were the first to optimize both QM (at the Hartree–Fock level) and MM regions simultaneously using link atoms.¹¹ Field *et al.* place the link atom 1 Å away from the QM boundary along the partitioned bond and optimize the system with restraints on the link atom position.¹² Hillier and co-workers use a conjugate gradient algorithm to minimize the MM region and a quasi-Newton algorithm minimizes the QM region.³⁹ They explored various sets of coupled constraints for performing QM/MM minimization.

There are several problems with the single link atom method. Some of the more egregious problems with the link atom approach result from leaving the MM valency at the severed bond effectively unsaturated. This can result in an unbalanced or arbitrarily redistributed charge. To alleviate

this problem, we introduce a new link atom method that involves the insertion of two extra atoms at the severed bond, one to saturate the QM fragment and the other to saturate the MM fragment. We call this method the double link atom (DLA) method to distinguish it from the single link atom (SLA) method in common use.

Problems of unbalanced electrostatic field or arbitrarily redistributed charges are not limited to link atom partitions. Frozen orbital, and pseudopotential methods of QM/MM partition also have severe difficulties with preserving electrostatic properties such as the dipole moment. To alleviate the problem, most methods neglect the electrostatic interaction of the MM host atom or the MM host group with the QM region, and this may have unrealistic effects in a condensed medium. The unbalanced electrostatic field along the partitioned bond could be due to the use of MM point charges for its interaction with the QM atoms. For atoms far from the QM region, the use of MM point charges is reasonable, but point charges could adversely modify electrostatic interactions between the QM region and the MM atoms immediately surrounding it. To address this matter we have implemented an approach, based on a suggestion by Logunov,⁴⁰ to “spread out” the charges of MM atoms near the QM/MM interface using a delocalized Gaussian function. We call this the delocalized Gaussian MM charge (DGMM) method. By thus empirically modeling the nonlocal character of electron charge distributions, a more consistent and realistic treatment of the immediate electrostatic environment of the QM region is made possible. Improvements in some properties result and are described below. We use the acronym PMM (point molecular mechanics) to refer to the more standard method, in which point charges for MM atoms are used in the QM portion of the calculation.

Note that the double link atom and single link atom methods can be combined with either the point charge (PMM) or the delocalized Gaussian (DGMM) method. We indicate a particular combination with a slashed notation. For example, DLA/DGMM indicates that the double link atom method is used and the Gaussian blur of MM charges interacts with QM atoms. We evaluate the DLA and DGMM methods for a series of model systems designed to mimic properties important for enzyme catalysis. Proton transfers are ubiquitous in enzyme catalysis and must be modeled accurately. Treatment of the charges of MM atoms near the QM/MM boundary is critical, as proton affinities are sensitive to electrostatics. We calculate proton affinities and deprotonation energies for a number of QM/MM protocols, using a variety of molecules with different functional groups. Defining a QM region of reasonable size in an enzyme may require link atoms to be positioned near each other. Therefore, we evaluate rotational barriers for butane, and the protonation energies of isopropyl alcohol and t-butyl alcohol with various QM/MM partitioning schemes. Some have more than one QM/MM partitions. Results for the SLA/PMM and DLA/DGMM approaches are compared. The best results are obtained using the combined DLA/DGMM approach.

Our QM/MM implementation with the DLA and the DGMM approaches are described in Sec. II followed by a

justification in Sec. III. The calculations are presented in Sec. IV and the findings are summarized in Sec. V.

II. METHODS

A. General approach

Our combined QM/MM potential is based on an earlier proposal of Field *et al.*,¹² but with significant modifications described below. In a QM/MM scheme, the chemical system is divided into two regions. Interactions are calculated with a classical parametrization within one region (MM region) and with a quantum mechanical method in the other (the QM region). In addition, a method for interfacing the two regions must be devised. Further terms describing boundary effects and restraints can be included and are well described in the classical simulation literature.^{4,41}

Our total QM/MM Hamiltonian H_{eff} is thus given by the sum

$$H_{\text{eff}} = H_{\text{QM}} + H_{\text{MM}} + H_{\text{QM/MM}}, \quad (1)$$

where H_{QM} and H_{MM} describe interactions within the quantum mechanical and classical regions, respectively, and $H_{\text{QM/MM}}$ is an interface term.

The $H_{\text{QM/MM}}$ Hamiltonian is the sum of the first three terms in Eq. (2). The fourth term H_{linear} is a linear restraint term and is only used when linear constraints are not employed

$$H_{\text{QM/MM}} = H_{\text{bonded}} + H_{\text{elec}} + H_{\text{vdw}} + H_{\text{linear}}. \quad (2)$$

Some classical energy terms for interactions between MM and QM atoms near the interface must be included in order to maintain reasonable geometries. These are described by H_{bonded} and comprise two-center bonds, three-center angles, four-center dihedrals, and four-center improper dihedrals. As described earlier, several groups include all classical bond terms between MM and QM atoms, but this has the drawback that some classical energy terms may duplicate interactions implicitly treated by the QM calculations. To avoid such duplication, we have devised a new set of rules for determining which bonded interactions between atoms at the QM/MM interface to include in the classical treatment. These rules, and a succinct and consistent terminology for the pertinent atoms are presented in Tables I and II. The rules are summarized by: (1) All mixed (involving both MM and QM atoms) bond terms are included. (2) Mixed angle terms are included only if the central atom is MM. (3) Mixed dihedral angle terms are included only if one or both central atoms are MM. (4) Mixed improper dihedral angle terms are included only if the central atom is MM.

Classical van der Waals terms for interactions between QM and MM atoms are necessary so that the QM region can respond to the sizes of the MM atoms. This is accomplished by assigning classical van der Waals parameters to the QM atoms, the most widely used procedure. The quantum mechanical link atom, however, has no van der Waals interactions with MM atoms.

We examined two approaches to cap the electron density along the partitioned bond: the commonly used single link atom method; and a new method, the double link atom

TABLE I. Atom type abbreviations.

Atom type	Abbreviation	Description
MM Atom	MMA	Any atom within the classical region.
QM Atom	QMA	Any atom within the quantum region.
QM Link atom	QML	An atom added to the QM region which serves to cap the electron density when a QM/MM boundary bisects a covalent QMA–MMA bond. The link atom should not distort the MM region, nor introduce large interactions in the QM region.
MM Link atom	MML	An atom added to the MM region which serves to cap the MM part of the system when a QM/MM boundary bisects a covalent QMA–MMA bond.
MM Host	MMH	The MM boundary atom of the bisected QMA–MMA bond.
QM Host	QMH	The QM boundary atom of the bisected QMA–MMA bond.

method described in Sec. II B. In the single link atom approach, a “link atom” is inserted along the bond vector severed by the QM/MM interface in order to terminate the quantum mechanical electron density. We call the MM and QM atoms joined by the severed bond the molecular mechanical host (MMH) and quantum mechanical host (QMH), respectively. We label the link atom QML (for quantum mechanical link atom) to distinguish it from another kind of link atom introduced in the double link atom method. The link atom is treated as a hydrogen atom, although in principle it could be any type of atom or a group of atoms. The only interactions of the QM link atom with the MM region are via the QM/MM electrostatics and via the optional restraint term, H_{linear} , included to maintain the link atom’s position along the QMH–MMH bond vector during minimization. H_{linear} is the classical angle parameter corresponding to the QML–QMH–MMH bond angle and has an optimal angle of 0° and a stiff force constant. The preferred approach to preserve linearity during minimization and dynamics involves constraints on the link atoms using the lone pair facility of CHARMM. The lone pair facility sets the mass of the link atoms to zero, and facilitates dynamics calculations without the need for any additional rigid body approximations. Minimizations with constrained link atoms produces superior geometries and energies compared to minimization using the H_{linear} restraint term. There are three different choices for constraining the link atom during minimization.

- (1) Fixed distance: The QML is placed at an appropriate fixed distance from the QMH, along the QMH–MMH bond vector.
- (2) Fixed ratio: The QML is placed along the QMH–MMH bond vector such that the ratio of the QMH–QML distance to the QMH–MMH distance is a constant.
- (3) Flexible: The QML remains collinear along the QMH–MMH bond, but its distance from the QMH varies based on the Hamiltonian.

For most of the results presented here, the fixed distance approach is used.

TABLE II. Classically computed boundary interactions.

Term	# Atoms	Classically computed when:
Bonded terms (involving QMA and MMA)		
Bond	2	QMA–MMA
Angle	3	X–MMA–X
Dihedral	4	X–MMA–X–X or X–X–MMA–X
Improper	4	MMA–X–X–X, central atom is in first position
Bonded terms (involving link atoms)		
Bond	2	MML–MMH bond calculated for double link atom scheme
Angle	3	(a) QML–QMH–MMH: Linear restraint added to maintain QML position along the QMH–MMH bond vector. (b) MML–MMH–QMH: Linear restraint added to maintain MML position along the MMH–QMH bond vector. These linear restraints are not needed if link atoms are constrained.
Nonbonded terms (involving QMA and MMA)		
van der Waals	2	QMA–MMA
Electrostatic	2	Never calculated classically. MMA atoms are treated as point or blurred charges and resultant electrostatic field interacts with the QMAs through the one-electron integrals. In the SLA method, the MMH is excluded from the electrostatic field. In the SLA/EXGR method (denoted EXGR only), all MM atoms belonging to MMH's neutral group are excluded.
Nonbonded terms (involving link atoms)		
van der Waals	2	Never: QMLs should be invisible to MM region. MML is invisible to both MM and QM atoms.
Electrostatic	2	QML, like other QM atoms, feels the MMA electrostatic field through its one-electron integrals. MML has electrostatic interactions with other MM atoms.

No other classical energy terms involving link atoms are computed. We emphasize that link atoms are not intrinsic components of the system being modeled and should not influence the MM region.

To account for the electrostatic interactions between QM and MM atoms, most of the MM atoms are treated as external charges interacting with the QM atoms through the one-electron integrals. In the “standard” PMM implementation, the MM atoms are treated as point charges. In the DGMM method mentioned earlier, Gaussian functions are used to “spread out” the charges of MM atoms near the QM/MM interface. Our implementation of this method is described in Sec. II C.

B. Double link atoms

We introduce a new method, called the double link atom (DLA) method, for treating bonds severed by the QM/MM boundary. In this method, two link atoms are inserted along each severed bond. One of the link atoms belongs to the QM region and is a QML, treated as described above. The other link atom belongs to the MM region and is called an MM link atom (MML). The MML is bonded to the MMH along the MMH–QMH bond vector. In the MM region, the MML behaves as a standard hydrogen. All classical bonded and

electrostatic energy terms involving the MML and MM atoms are included. But, classical bonded terms involving the MML and QM atoms are not calculated. The MML does not participate in any van der Waals interactions with QM atoms, as this would duplicate the QMH van der Waals interactions. During minimization, MML is constrained along the MMH–QMH bond vector with the same approach used for constraining the quantum mechanical link atom.

The double link atom method can be explained by the following two steps also shown in Fig. 1: (1) Partition a large MM molecule into smaller molecules which overlap, so they preserve properties such as net charge, dipole moment, etc. (2) Convert one of the smaller molecules from MM to QM (shaded region in Fig. 1). Thus, partitioning a chemical system with double link atoms creates “complete molecules,” some treated classically, with appropriate charges and atom types, and some quantum mechanically. Single link atom insertion creates a “complete” QM molecule, but a fragmented MM region that may differ greatly from the overall system in net charge and dipole, unless other adjustments are made.

C. Delocalized Gaussian MM charges

Delocalized Gaussian functions can be used to modulate the electrostatic interaction of MM atoms with the QM re-

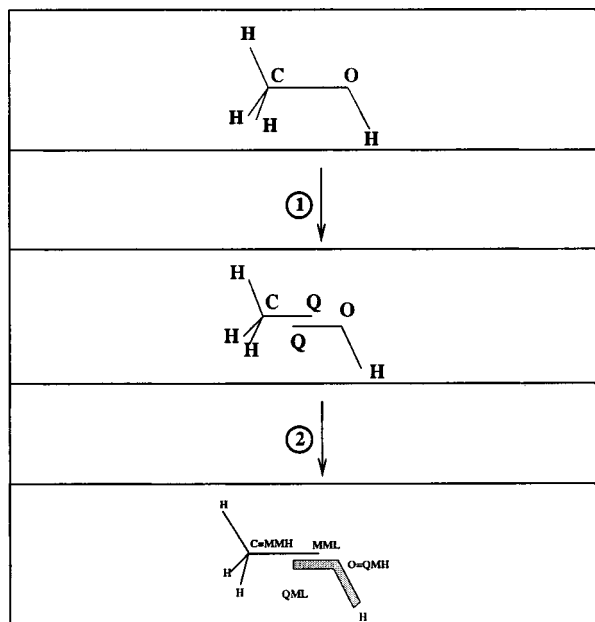


FIG. 1. Partitioning of methanol molecule to explain the double link atom method. The C–O bond is partitioned. “Q” represents the new link atom positions. QMH and QML represent the QM host atom and the QM link atom, respectively. MMH and MML represent the MM host atom and the MM link atom, respectively.

gion. The principle is that a delocalized charge distribution better mimics the real charge distribution of an atom than does a point charge, and the difference may be important for atoms near the QM/MM interface. The delocalized Gaussian function (χ_{DG}) is a simplified form of the general basis function with exponent α centered at the point $R = (R_x, R_y, R_z)$ used in *ab initio* methods and given by

$$\chi(R, \alpha, l, m, n) = \left(\frac{2\alpha}{\pi} \right)^{3/4} \frac{(4\alpha^{(l+m+n)/2})}{[(2l-1)!(2m-1)!(2n-1)!]^{1/2}} \times x^l y^m z^n e^{-\alpha(r-R)^2}. \quad (3)$$

From the above we keep only the Gaussian function ($l = m = n = 0$)

$$\chi_{\text{DG}}(R, \alpha) = \left(\frac{2\alpha}{\pi} \right)^{3/2} e^{-\alpha(r-R)^2}. \quad (4)$$

The contribution of the delocalized charges to an element of the Fock matrix F_{ij} is given by the following summation over all DGMM centers:

$$F_{ij} = \sum_k \int \int dr_1 dr_2 \chi(R_i, \alpha_i, l_i, m_i, n_i) \times \chi(R_j, \alpha_j, l_j, m_j, n_j) \frac{1}{r_{12}} \chi_{\text{DG}}(R_k, \alpha_k). \quad (5)$$

Since a Gaussian function can be expanded as the product of two Gaussian functions with a common origin and half the exponent, we can express the integrals required as follows:

$$F_{ij} = \sum_k \int \int dr_1 dr_2 \chi(R_i, \alpha_i, l_i, m_i, n_i) \times \chi(R_j, \alpha_j, l_j, m_j, n_j) \times \frac{1}{r_{12}} \chi\left(R_k, \frac{\alpha_k}{2}, 0, 0, 0\right) \chi\left(R_k, \frac{\alpha_k}{2}, 0, 0, 0\right), \quad (6)$$

which can be readily computed using a standard four-center integral code.

Interaction between the delocalized MM charge (i) with charge q_i and the QM nucleus (j) with charge Z_j at the distance r_{ij} is:

$$E_{ij} = \frac{q_i Z_j}{r_{ij}} \operatorname{erf}\left(\frac{r_{ij}}{\sigma}\right), \quad (7)$$

where σ is the Gaussian blur width. As $\alpha \rightarrow \infty$ in Eq. (4), Eq. (5) becomes the nuclear attraction integral used in standard point charge methods. Interaction between the delocalized charge and any QM atom is always finite even in the limit when the distance between the two is zero. This property allows inclusion of both the MML and the MMH atoms in the QM/MM potential, simplifying the rules. Besides spherically symmetric delocalization, other forms of charge distribution can also be handled. The delocalized charge function can be applied to any MM atom because the current implementation does not significantly increase the CPU time usage, but as might be expected, remote charges act just as point charges when the distance is considerably larger than σ . The DGMM method in the CHARMM/GAMESS-UK⁴² and CHARMM/GAMESS-US⁴³ interface software is implemented in a flexible way so that one can mix point charges with DGMM charges in the same run and each of the DGMM charges may have its own Gaussian width value.

Differences between the single and double link atom approaches will be important in systems with large or multiple classical regions. This study employs small molecules in the gas phase and is not greatly influenced by the single link atom deficiencies. However, if such small molecules were solvated, these issues would become much more important. The double link atom method is also important in the proton affinity studies, where it is able to reproduce all-QM results almost exactly. The double link atom method should be used with the Gaussian blur of MM charges. Use of the double link atom method with point MM charges is not recommended. But, MM atoms very far from the QM atoms may have negligible effect on blurring. When MM charges are blurred with the DGMM method, the MML and the MMH both contribute to the electrostatic interaction. In our calculations, all MM atoms are assigned the same value of σ .

Electrostatic interactions between MM atoms can be treated with or without a cutoff distance or by an Ewald sum. In our implementation with the current test suite, no cutoffs or electrostatic scaling were used, either for MM electrostatics or for QM/MM electrostatic interactions.

The molecular simulation program CHARMM²⁴ was interfaced with the quantum mechanical package GAMESS-UK⁴² to provide a broad range of *ab initio* methods for the quantum mechanical portion of the QM/MM calculations. Classical energy parameters were taken from CHARMM22 parameter

sets.^{44,45} In the CHARMM/GAMESS-UK implementation, MD and energy minimization, with constraints on the link atoms, are performed by the MM side of the program. QM energies and forces are calculated in the presence of the electrostatic field for each minimization step. The forces exerted by and on the QM atoms are returned to CHARMM. A classical energy and force calculation is performed and the QM and MM forces are combined. The combined set of forces determines the structural change using standard techniques (MD, energy minimization, etc.). Quantum calculations were performed at HF and DFT levels with different split valence basis sets. Becke's three-parameter hybrid functional,^{46,47} B3LYP, as implemented in GAMESS-UK was used for DFT calculations. A word about the notation used in the discussion. Double link atom calculations were always performed with Gaussian blur of MM charges and are denoted by DLA/B, where B is the Gaussian blur width in angstroms. It should be noted that a Gaussian blur is used to blur MM charges for their interactions with QM atoms, but point charges are still used to calculate electrostatic interactions between MM atoms. Single link atom calculations are denoted SLA or EXGR depending on how QM/MM electrostatics is handled. Single link atom calculations performed with exclusion of the MMH atom charge from the QM calculation, and point charges on other MM atoms, are denoted by SLA. Single link atom calculations performed with exclusion of the entire MM host group from the QM calculation, and point charges on other MM atoms are referred to by EXGR. Calculations were performed on LOBOS, a large Beowulf-class computer cluster.⁴⁸ An interface of our implementation with GAMESS-US⁴³ is also available.

III. JUSTIFICATION

The development of the DLA and DGMM methods are based on the following principles:

- (1) There should be conservation of charge (charge consistency), i.e., charge should neither be created nor destroyed at an interface.
- (2) There should be minimal perturbation of dipole and electrostatic distribution.
- (3) The method should work well at any level of QM theory.
- (4) Local deformation energy should be approximately constant. There should be no overcounting or undercounting of local energy terms.
- (5) The method should work well with multiple interface regions in close proximity.

The single link atom partitioning schemes have intrinsic errors in the calculation of dipole and quadrupole moments, and can create polarization along the partitioned bond. Polarization can produce significant errors in the calculation of deprotonation and protonation energies. We have computed QM/MM dipole moments of ethane and butane. Ethane is partitioned along the central C—C bond. Single link atom partition creates a QM methane and a MM CH₃ group. The SLA dipole moment along the x axis is -1.38 debye (it has no significant dipoles along the other axes). This shows the polarization induced by putting a single link atom to saturate

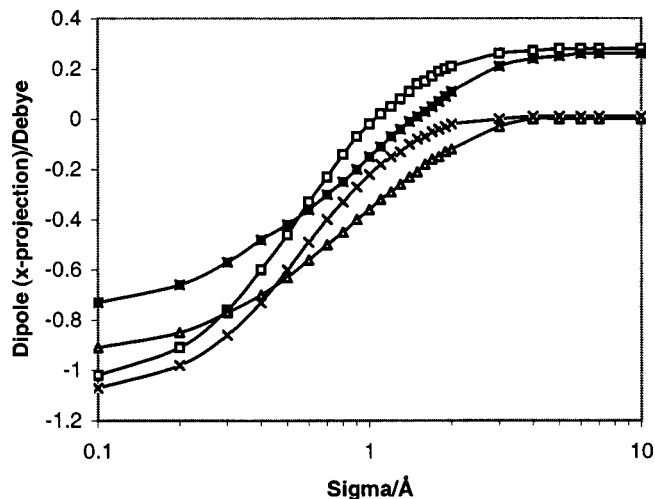


FIG. 2. QM/MM dipole moment of ethane and butane as a function of the Gaussian width, with the double link atom method. The x component of the dipole is analyzed for both molecules. The QM level is RHF/6-31G*. X: Ethane, standard CHARMM charges; $\square$: Ethane, moving 0.05 charge units from MML to MMH; $\triangle$: Butane, standard CHARMM charges; $\blacksquare$: Butane, moving 0.05 charge units from MML to MMH.

the quantum region only. Double link atom partition of ethane creates a QM methane and a MM methane. Figure 2 shows the QM/MM dipole moment of ethane, partitioned with the double link atom method, at different blur widths (RHF/6-31G* level). At a blur of 1 Å, the dipole is -0.25 debye. Thus, DLA produces a much better electrostatic representation than does SLA. In the single link atom, by excluding the electrostatic interaction of the MM host group (i.e., MM methyl group) with the QM methane (the EXGR method), the dipole moment is -0.46 D. The polarization when electrostatic interaction of the entire host group is neglected is not as strong as the one produced by neglecting the electrostatic interaction of the MM host atom only, but is still larger than the polarization produced by DLA. This explains why reasonable protonation and deprotonation energies can be obtained in many cases by neglecting the electrostatic interaction of the entire host group. Single link atom partition of butane forms a QM ethane and a MM CH₃CH₂. At RHF/6-31G* level, the x component of the SLA and EXGR dipole moments of butane are -1.36 and -0.85 D, respectively. DLA partition forms a MM ethane. DLA with σ between 1 and 2 Å has QM/MM dipole between -0.36 and -0.12 D. Similar results are obtained at B3LYP/6-31G* level. It indicates DLA protonation and deprotonation energies between blurs of 1 and 2 Å should provide robust results.

In the MM methane (DLA case), the carbon has a charge of -0.36 and the link atom and hydrogens have charges of 0.09 units (standard CHARMM charges). One may obtain zero DLA dipole moment by moving a small partial charge from the MM link hydrogen atom to the MM host carbon. The charge to be moved depends on the polarization induced, and hence on the Gaussian blur width. Figure 2 indicates that a charge shift of 0.05 units from the standard CHARMM charges can produce zero DLA dipole moment at blur of 1 Å. This results in a charge of -0.41 on the MMH carbon, and a

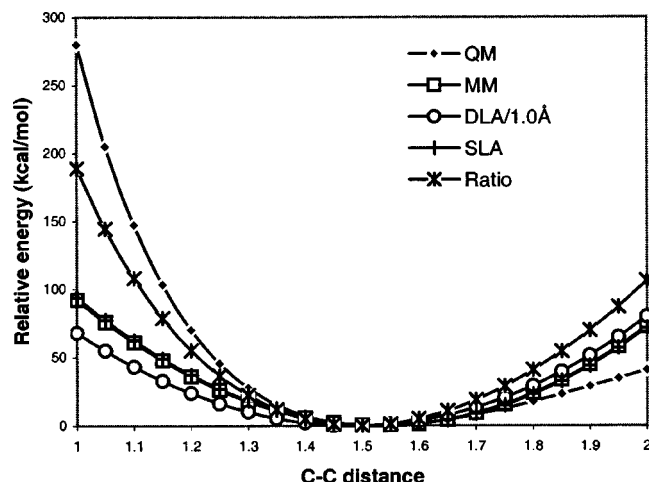


FIG. 3. Ethane stretch potential by various methods. The QM level is at RHF/6-31G*. Ratio is obtained by setting the QML-QMH to MMH-QMH distance to 0.727.

charge of 0.14 on the MML hydrogen. Similar results are obtained at RHF/3-21G level. Figure 2 also illustrates the effect of shifting standard CHARMM charges on the DLA partition of butane. Zero DLA dipole moment of butane at a Gaussian width of 1.5 Å can be obtained with a charge shift of 0.05 units. The shift of charge between the MMH and the MML is a free parameter which may be used to fine-tune total dipole moments or other observables such as proton affinities. For the results presented in this paper, a shift of zero (i.e., no shift) has been used in order to demonstrate the quality of results in absence of this fine tuning.

Questions have been raised as to whether a smooth potential energy surface can be obtained by using link atoms.^{15,35} In Thiel's mechanical embedding scheme, there is a chance that atoms might collapse while performing minimization. The potential energy calculated by QM, MM, DLA, and SLA methods for ethane is shown in Fig. 3. The QM methyl group is held rigid and the MM methyl group is translated along the C-C bond, with the C-C distance varying from 1 to 2 Å. SLA and DLA/1 Å calculations were performed by placing the link atoms 1.111 Å away from their respective hosts. The potential energy was also calculated by placing the QML such that the ratio of QML-QMH to MMH-QMH distance is 0.727. The DLA/1 Å, C-C equilibrium distance is about 0.05 Å shorter than the corresponding MM distance. The use of constraints with the SLA method results in a harmonic well with a limited probability of collapse. The use of DLA/DGMM removes any possibility of such a collapse in that forces go to zero as MM atoms fully overlap with QM atoms.

IV. RESULTS AND DISCUSSION

A. Rotational barriers: Butane

Butane was partitioned into MM and QM domains, using link atoms for each domain junction. Three two-domain and two three-domain partitioning schemes were studied

Model

1	CH₃ -CH ₂ -CH ₂ -CH ₃
2	CH₃ - CH₂ -CH ₂ -CH ₃
3	CH₃ - CH₂ - CH₂ -CH ₃
4	CH₃ -CH ₂ -CH ₂ - CH₃
5	CH ₃ - CH₂ - CH₂ -CH ₃ .

Quantum regions are indicated with boldface, underlined text. For example, in model 1 a link atom is placed along the C1-C2 bond. C1 is the quantum carbon, and C2-C3-C4 are MM carbons. Double link atoms and single link atoms were employed with the five butane models at the RHF/6-31G* level. Model 5 was also studied at the B3LYP/6-31G* level.

The barrier to rotation about the central C-C bond was calculated for each partitioning scheme. This was done by rotating the C1-C2-C3-C4 dihedral angle, ψ , from 180° (*trans* butane) to 0° (*cis* butane) at 10° intervals and energy minimizing each geometry. During minimization, the angle ψ was maintained with a harmonic restraint having a force constant of 200 kcal/mol/radian². Barriers were calculated using both RHF/6-31G* and B3LYP/6-31G* levels of theory in the QM portion of the calculations. Results for the five QM/MM schemes and for pure MM and pure QM calculations are presented in Table III. The rotational barrier for model 5 is shown in Fig. 4. The energy of the *trans* orientation, which corresponds to 0°, was set to 0.0 kcal/mol in every case.

B3LYP/6-31G* and MM rotational energies using the CHARMM potential function agree within 0.1 kcal/mol for ψ , from 40° to 170°, and differ by only 0.4 kcal/mol at 0°. The *trans*-*cis* barrier at B3LYP/6-31G* is 5.6 kcal/mol and the MM barrier is 5.2 kcal/mol. The RHF/6-31G* *trans*-*cis* barrier (of 6.1 kcal/mol) is higher than the MM barrier by 0.9 kcal/mol, but the barriers vary by a maximum of 0.3 kcal/mol for ψ , from 40° to 170°.

QM/MM rotational barriers were calculated using single link atom and double link atom with Gaussian blur schemes. For double link atom methods, the best overall rotational profile is obtained with a σ of 1 Å.

Model 5 clearly shows the improvement achieved by double link atoms over single link atoms. It also highlights our treatment of handling multiple link atoms if they are placed close to each other (1,4 for model 5). The SLA energy profile for model 5 is significantly lower than the energy profile obtained with QM or MM treatment alone. At $\psi=0^\circ$, deviations from the MM energies are 2.1 and 2.4 kcal/mol, at RHF/6-31G* and B3LYP/6-31G* levels, respectively. SLA energies are actually lower for the *gauche* conformers, with ψ between 50° and 80°, than for the *trans* conformation. The *gauche* conformer is lower by about 0.7 kcal/mol at $\psi=60^\circ$, at both RHF/6-31G* and B3LYP/6-31G* levels. The discrepancies are principally due to neglect of the electrostatic interaction between the MMH and atoms in the QM region. The CHARMM force field, where nonpolar hydrogens have a typical partial charge of 0.09, exhibits this behavior. Methods such as MM3 that employ no charges will not encounter this problem. The rotational barrier profile produced by double link atoms with Gaussian blur shows significant improvement over SLA treatment (Fig. 4). A Gaussian blur of 1.5 Å produces a better profile than a blur of 0.5 Å. Due to the

TABLE III. Butane rotational barrier.

Dihedral angle	MM	QM ^a	Model 1 ^a		Model 2 ^a		Model 3 ^a		Model 4 ^a		Model 5 ^a		QM ^b	Model 5 ^b	
			SLA	DLA	SLA	DLA	SLA	DLA	SLA	DLA	SLA	DLA		SLA	DLA
0	5.2	6.1	5.2	4.9	4.4	4.7	4.5	4.6	8.6	8.4	3.1	5.1	5.6	2.8	5.2
10	4.9	5.8	4.8	4.6	4.1	4.3	4.2	4.3	8.2	8.0	2.8	4.9	5.3	2.6	4.9
20	4.1	4.8	4.0	3.8	3.4	3.6	3.5	3.6	7.2	7.1	2.1	4.2	4.4	1.9	4.2
30	3.1	3.6	3.0	2.8	2.4	2.5	2.5	2.6	6.1	5.9	1.2	3.2	3.3	1.1	3.3
40	2.1	2.4	2.1	1.9	1.5	1.6	1.5	1.6	4.9	4.7	0.3	2.3	2.1	0.2	2.4
50	1.3	1.4	1.3	1.1	0.8	0.8	0.7	0.8	4.0	3.7	-0.4	1.5	1.3	-0.4	1.6
60	0.9	1.0	0.9	0.7	0.4	0.4	0.3	0.4	3.3	3.1	-0.7	1.0	0.9	-0.8	1.1
70	0.8	0.9	0.8	0.7	0.4	0.4	0.3	0.4	3.0	2.8	-0.7	0.9	0.9	-0.7	1.0
80	1.2	1.3	1.2	1.0	0.8	0.8	0.8	0.9	3.0	2.8	-0.2	1.2	1.2	-0.2	1.2
90	1.8	2.0	1.8	1.7	1.6	1.6	1.6	1.6	3.2	3.1	0.6	1.8	1.9	0.5	1.8
100	2.6	2.8	2.6	2.5	2.4	2.4	2.5	2.5	3.7	3.6	1.5	2.5	2.6	1.4	2.5
110	3.2	3.4	3.2	3.2	3.1	3.1	3.1	3.1	4.0	4.0	2.3	3.0	3.1	2.2	2.9
120	3.4	3.6	3.4	3.5	3.4	3.4	3.4	3.3	4.0	4.1	2.7	3.2	3.3	2.5	3.1
130	3.1	3.3	3.1	3.2	3.1	3.2	3.2	3.1	3.4	3.6	2.7	2.9	3.0	2.5	2.8
140	2.4	2.6	2.4	2.4	2.4	2.4	2.5	2.4	2.6	2.7	2.2	2.3	2.4	2.0	2.2
150	1.5	1.7	1.5	1.5	1.5	1.5	1.6	1.6	1.6	1.7	1.4	1.5	1.6	1.3	1.4
160	0.7	0.8	0.7	0.7	0.7	0.7	0.8	0.8	0.7	0.8	0.7	0.7	0.8	0.6	0.7
170	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2

^aAt RHF/6-31G* level. SLA is the single link atom calculation in which the electrostatic interaction of the MM host atom is neglected with the QM atoms, DLA is with $\sigma=1$ Å.

^bAt B3LYP/6-31G* level. DLA is with $\sigma=1.5$ Å.

presence of two QM/MM partitions in this model, the MM link atoms are positioned 1–4 to each other. Nonbonded (electrostatic and van der Waals) interactions between MM link atoms positioned 1–3 or 1–4 to each other are neglected to avoid spurious short-range interactions. This is achieved by forming pseudobonds between MM link atoms. Figure 4 also shows the rotational barrier profile when 1–4 interactions between the MM link atoms are not neglected. By ne-

glecting the short-range nonbonded interactions, the rotational barrier profile better matches the QM and MM profiles.

Model 1 with a large MM region, and one QM/MM interface region, produces a good profile for the DLA with a 1 Å σ (hereafter denoted DLA/1Å). The SLA rotational barrier profile is also reasonable.

Model 2 and model 3 barriers for SLA and DLA methods are similar. For ψ from 0° to 40°, DLA barriers are in better agreement with the MM barriers than are the SLA barriers. For ψ from 50° to 80°, SLA and DLA barriers are almost identical, but they underestimate from the MM barrier by about 0.5 kcal/mol. MM, SLA, and DLA barriers are almost identical for ψ from 100° to 170°.

Model 4 has an energy profile significantly higher than that for the pure MM or QM treatments. At $\psi=0^\circ$, the DLA and SLA barriers are higher than the MM barrier by 3 and 3.4 kcal/mol, respectively. The exaggerated barriers may be traced to the steric repulsion of two QM methyl groups brought into close proximity. Partitioning of a QM/MM region when two QM groups are placed very close to each other should be avoided.

B. Proton affinities

Modeling enzyme catalysis requires that the energetic cost of proton exchange be predicted accurately. Thus, QM/MM calculations must accurately reproduce proton affinities for groups that participate in proton exchange. Proton affinity may be defined as the energy released (denoted with negative values) when the neutral molecule accepts a proton to form the protonated molecule. Proton affinities were calculated by rigorously energy minimizing the neutral molecule, adding a proton to the appropriate atom (O or N), and reminimizing the energy. The difference between the mini-

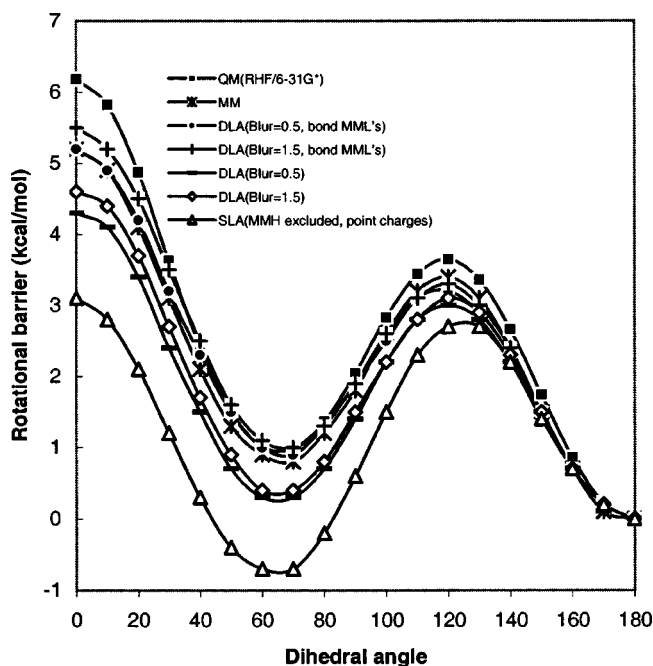


FIG. 4. Butane (C1–C2–C3–C4) rotational barrier for model 5. C2–C3 is the QM region. C1 and C4 are treated by MM. The QM level is at RHF/6-31G*.

TABLE IV. RHF Protonation energies.

Molecule	QM Level	QM	SLA ^a	EXGR ^b	DLA ^c /1 Å	DLA/999 Å	DLA/ 0.001 Å
<chem>CH3-CH2OH</chem>	RHF/3-21G	-209.5	-173.0	-204.9	-208.6	-204.9	-212.8
<chem>CH3-CH(CH3)OH</chem>	RHF/3-21G	-213.2	-178.7	-209.0	-212.2	-209.0	-217.7
<chem>CH3-C(CH3)2OH</chem>	RHF/3-21G	-213.2	-140.2	-204.5	-211.6	-204.5	-222.2
<chem>CH3-C(CH3)3</chem>	RHF/3-21G	-216.1	-183.0	-213.4	-216.1	-213.4	-219.0
<chem>CH3-C(CH3)2COH</chem>	RHF/3-21G	-216.1	-145.7	-208.4	-215.0	-208.3	-224.2
<chem>CH3-C(CH3)3</chem>	RHF/3-21G	-216.1	-110.2	-204.5	-214.5	-204.4	-230.8
<chem>H3C-H2C-CH2-CH2-OH</chem>	RHF/6-31G*	-197.8	-179.9	-197.4	-197.4	-194.9	-200.7
<chem>H3C-CH2-NH2</chem>	RHF/3-21G	-240.1	-204.7	-237.0	-239.9	-237.0	-244.6
<chem>HO-H2C-CH2-CH2-OH</chem>	RHF/6-31G*	-195.7	-168.6	-191.5	-192.4	-190.1	-194.8
<chem>HO-CH2-CH2-CH2-NH2</chem>	RHF/6-31G*	-231.8	-207.1	-229.5	-229.7	-228.2	-232.3
<chem>H2N-CH2-CH2-CH2-NH2</chem>	RHF/6-31G*	-233.2	-206.4	-228.8	-229.0	-228.2	-228.2
<chem>H2N-C(=O)-CH2-CH2-C(=O)-NH2</chem>	RHF/6-31G*	-201.9	-181.8	-200.1	-201.1	-200.0	-203.2
<chem>C1CCCCC1N</chem>	RHF/6-31G*	-236.8	-198.3	-235.8	-236.5	-231.1	-242.0

^aSLA is the single link atom method in which electrostatic interaction of MM host atom is neglected with QM atoms.

^bEXGR is the single link atom method in which electrostatic interaction of the MM host group is neglected with QM atoms.

^cDLA/X is the double link atom method where X is the Gaussian blur width.

mized energies of the two species gives the proton affinity. Zero-point energy corrections are not calculated. The proper test of a QM/MM method is its ability to reproduce the results of QM calculations at any given level of theory. The proton affinity of several molecules is examined. Some of the molecules in the test suite have also been used by others^{15,16} in their QM/MM studies. The results are presented in Table IV for RHF and in Table V for DFT calculations.

The proton affinity of ethanol, isopropyl alcohol, and t-butyl alcohol is examined at RHF/3-21G and B3LYP/6-31G* levels. Isopropyl alcohol is treated with one and two partitions, and t-butyl alcohol is treated with one, two, and three partitions. T-butyl alcohol with three DLA partitions corresponds to three methanes treated by MM and one methanol treated by QM, plus their interactions. At the B3LYP/6-31G* level, the proton affinities (PA) follow the trend

$$PA_{\text{ethanol}} < PA_{\text{isopropyl alcohol}} < PA_{\text{t-butyl alcohol}}.$$

T-butyl alcohol has the highest proton affinity (more energy is released) because the resulting cation can be stabilized better by three methyl groups than the two methyl groups of isopropyl alcohol or the single methyl group of ethanol.

SLA treatment can grossly underestimate proton affinities. If multiple interface regions are used for isopropyl alcohol and t-butyl alcohol, the trend in proton affinities is not correctly reproduced. The SLA proton affinity of ethanol varies from the B3LYP/6-31G* proton affinity by 38.4 kcal/mol. For isopropyl alcohol, the SLA proton affinities vary from the QM value by 36.4 and 75.7 kcal/mol for one QM/MM interface and two QM/MM interfaces, respectively, at the B3LYP/6-31G* level. For t-butyl alcohol, the SLA proton affinities differ from the QM values by 35.2, 73.1, and 111.3 kcal/mol for one, two, and three QM/MM interfaces, respectively, at the B3LYP/6-31G* level. It is clear the SLA treatment cannot reproduce accurate proton affinities, and for interface regions in close proximity the results are totally

TABLE V. B3LYP protonation energies.

Molecule	QM Level	QM PA	SLA ^a	EXGR ^b	DLA ^c /1 Å	DLA/999 Å	DLA/0.001 Å
<chem>CH3-CH2OH</chem>	B3LYP/6-31G*	-194.9	-156.5	-188.9	-192.3	-188.9	-196.9
<chem>CH3-CH(OH)CH3</chem>	B3LYP/6-31G*	-199.6	-163.2	-194.1	-197.5	-194.1	-202.9
<chem>CH3-C(OH)(CH3)2</chem>	B3LYP/6-31G*	-199.6	-123.9	-188.6	-195.4	-188.5	-206.2
<chem>CH3-C(OH)(CH3)3</chem>	B3LYP/6-31G*	-203.9	-168.7	-199.0	-202.0	-199.0	-205.5
<chem>CH3-C(OH)(CH3)2CH2CH3</chem>	B3LYP/6-31G*	-203.9	-130.8	-193.6	-200.0	-193.5	-209.5
<chem>CH3-C(OH)(CH3)2CH2CH2CH3</chem>	B3LYP/6-31G*	-203.9	-92.6	-188.9	-198.9	-188.8	-214.7
<chem>H3C-CH2-CH2-CH2-OH</chem>	B3LYP/6-31G*	-197.5	-179.3	-197.0	-197.0	-194.4	-200.2
<chem>H3C-CH2-NH2</chem>	B3LYP/6-31G	-234.7	-198.6	-231.3	-234.0	-231.3	-238.5
<chem>HO-CH2-CH2-CH2-OH</chem>	B3LYP/6-31G*	-195.4	-167.4	-190.2	-191.0	-188.9	-193.3
<chem>HO-CH2-CH2-CH2-NH2</chem>	B3LYP/6-31G*	-231.0	-206.0	-228.3	-228.5	-227.0	-230.9
<chem>H2N-CH2-CH2-CH2-NH2</chem>	B3LYP/6-31G*	-232.5	-205.3	-227.7	-227.9	-227.0	-230.4
<chem>H2N-C(=O)-CH2-CH2-C(=O)-NH2</chem>	B3LYP/6-31G*	-206.4	-183.9	-202.7	-203.8	-202.5	-205.9
<chem>C1CCCCC1N</chem>	B3LYP/6-31G*	-236.1	-197.5	-235.0	-235.7	-230.3	-240.9

^aSLA is the single link atom method in which electrostatic interaction of MM host atom is neglected with QM atoms.

^bEXGR is the single link atom method in which electrostatic interaction of the MM host group is neglected with QM atoms.

^cDLA/X is the double link atom method where X is the Gaussian blur width.

inadequate. Ethanol, isopropyl alcohol (two interface regions), and t-butyl alcohol (three interface regions) SLA proton affinities do not have the same trend as the QM proton affinities.

The EXGR option, which neglects the electrostatic interaction of the neutral MM host group with QM atoms, produces better results than the SLA when single partitions are used for isopropyl alcohol and t-butyl alcohol. However, with two partitions for isopropyl alcohol and three partitions for t-butyl alcohol, excluding MM host group electrostatic interaction produces the same proton affinities for ethanol, isopropyl alcohol, and t-butyl alcohol. This is because for all three molecules, excluding the MM host group corresponds to calculating the proton affinity of methanol. This shows the inadequacy of neglecting electrostatic interaction of the MM host group when multiple interface regions are present.

The proton affinities produced with the double link atom method at a Gaussian blur of 1 Å agree well with the QM (B3LYP/6-31G*) proton affinities. The results are also robust when multiple partitioning schemes are used, and the correct

trend of proton affinities is produced. The DLA with an infinite blur of MM atoms is essentially the same as excluding the electrostatic interaction of all MM atoms with QM atoms. For these alcohols, it produces results similar to those given by the exclude group option. Using double link atoms together with point MM charges overestimates the proton affinities. This shows the importance of including the electrostatic interaction of all MM atoms with the QM region, and blurring the MM charges for accurate proton affinities. The DLA with a σ of 1 Å produces accurate proton affinities at both B3LYP/6-31G* and RHF/3-21G levels.

Butanol is partitioned along the C2–C3 bond. With double link atoms, this gives an ethanol (QM region) and an ethane (MM region). The SLA proton affinity differs by 18.2 kcal/mol from the QM proton affinity at the B3LYP/6-31G* level. The DLA/1 Å proton affinity is within 0.5 kcal/mol.

The proton affinity of ethyl amine (CH3NH2 as QM region, and CH4 as MM region) calculated using DLA/1 Å differs from the QM value by 0.7 kcal/mol at B3LYP/6-31G level, and by 0.2 kcal/mol at RHF/3-21G level. The SLA

TABLE VI. Deprotonation energies.

Molecule	Level	QM	SLA ^a	EXGR ^b	DLA ^c /1.7 Å	DLA/999 Å	DLA/0.001 Å
$\text{H}_3\text{C}-\text{H}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$	RHF/6-31G*	404.1	389.6	407.8	406.5	405.2	410.6
$\text{H}_3\text{C}-\text{H}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$	B3LYP/6-31G*	397.4	382.9	401.9	400.6	399.1	404.7
$\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$	RHF/6-31G**	369.7	343.2	372.5	374.0	372.5	378.7
$\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$	B3LYP/6-31G	359.0	332.9	361.1	362.3	361.1	367.2
$\text{CH}_3-\text{CH}_2-\text{SH}$	RHF/6-31G	354.7	325.6	355.5	357.0	355.5	362.1
$\text{CH}_3-\text{CH}_2-\text{SH}$	B3LYP/6-31G	360.2	331.9	361.8	363.2	361.8	368.3
$\text{O}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$	RHF/6-31G*	434.8	402.4	422.3	435.1	366.5	437.5
$\text{O}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$	B3LYP/6-31G*	436.9	401.6		435.2	365.9	437.8

^aSLA is the single link atom method in which electrostatic interaction of MM host atom is neglected with QM atoms.

^bEXGR is the single link atom method in which electrostatic interaction of the MM host group is neglected with QM atoms.

^cDLA/X is the double link atom method where X is the Gaussian blur width.

proton affinities vary by more than 30 kcal/mol.

Proton affinities for three bifunctional molecules, $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$, $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{NH}_2$, and $\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ are presented. SLA proton affinities vary from their respective QM values by at least 25 kcal/mol. DLA/1 Å proton affinities are reasonable. The agreement is comparable to those obtained by Thiel without MM polarization. Thiel obtained better agreement with the QM proton affinities (at the AM1 level) by incorporating MM polarization for these molecules. DLA/1 Å proton affinities for an amide and cyclohexyl amine are in very good agreement with the QM proton affinities. SLA proton affinities are gross underestimates.

C. Deprotonation energies

Deprotonation energies were calculated in the same manner as proton affinities. The acidic proton was deleted, and the difference between the minimized energies of the deprotonated and neutral species was computed. Deprotonation energies of several molecules, without zero-point corrections, are presented at HF and DFT levels in Table VI. SLA deprotonation energies of butanol differ from the QM deprotonation energies by about 15 kcal/mol, whereas DLA/1.7 Å differs by 2.4 kcal/mol. With increasing σ , the deprotonation energy of butanol should approach the deprotonation energy of ethanol, and the calculated value does that. DLA/1.7 Å deprotonation energy of propanoic acid, even though much better than the SLA treatment, differs by more than 4 kcal/mol. Better accuracy might be achieved in larger molecules where the QM/MM partition is farther from the reaction center. Accurate deprotonation energies are also obtained for $\text{CH}_3\text{CH}_2\text{SH}$. For the deprotonation energies of butanol, propanoic acid, and $\text{CH}_3\text{CH}_2\text{SH}$, neglecting the charge of the MM host group produces reasonable results. Exclusion of the MM host group charge does not quite preserve electrostatic

properties along the partitioned bond, as illustrated by the dipole moment studies of ethane and butane, but does produce reasonable deprotonation energies in some cases. However, for deprotonation of succinate anion ($\text{OOCCH}_2\text{CH}_2\text{COOH}$), implementation of the exclude group option has practical problems. The MM host group for succinate does not have an overall neutral charge, and empirical charge adjustments have to be made by moving charge to the nearby group to make the host group neutral. DLA with a Gaussian blur produces robust results, and further charge manipulation is avoided. SLA deprotonation energy differs by more than 30 kcal/mol. Using non-neutral MM host group charge exclusion (without further charge manipulation) produces a deprotonation energy that is wrong by 12 kcal/mol at the RHF/6-31G* level. The double link atom method with a Gaussian blur produces extremely good deprotonation energies, and the deprotonation energy at a blur of 1.7 Å at RHF/6-31G* level is only 0.3 kcal/mol greater than the QM value. Deprotonation energy at an infinite σ (i.e., discarding the electrostatic perturbation of QM atoms by MM atoms) produces the deprotonation energy of acetic acid.

D. Link atom position

Minimization and molecular dynamics should not be performed without any restraints or constraints on the link atom positions. If a link atom becomes significantly noncollinear with the associated QM/MM bond, then the assumptions used to define the link atom will no longer be valid. Bonds with the link atoms replace the partitioned bond; thus, the host atom to link atom bond should remain collinear with the partitioned bond. Giving freedom to the link atoms would not serve this purpose. We have observed nonphysical geometries and energies when employing QM/MM link atom methods without any restraints or constraints. Minimization

with restraints using the H_{linear} term in the Hamiltonian [Eq. (2)] sometimes puts enormous forces on the atoms, and the geometry may distort to relieve the strain. No such problems can be encountered during minimization when constraints are used on the link atoms.

As discussed in Sec. II A, there are three ways to constrain a link to be collinear. For most calculations in this study, both quantum mechanical and molecular mechanical link atoms have been constrained along the C–C bond vector 1.111 Å from their respective hosts. This is the optimum C–H bond distance from the CHARMM parameter file.⁴⁴ However, placement of the link atom, within reasonable distances, should not have a pronounced effect on the protonation and deprotonation energies. Using DLA/1.0 Å, and constraining the link atoms to be 1.000 Å away from the host atoms, the protonation energy of butanol changes by 0.1 kcal/mol compared to constraining the links 1.111 Å away (RHF/6-31G* level). The protonation energy of butanol is examined across a range of blur widths, and the maximum variation when placing the link atom from 1.000 to 1.111 Å away from its host is 0.6 kcal/mol. The deprotonation energy of butanol, using DLA/1.7 Å is raised by 1 kcal/mol when the link atoms are placed 1 Å from the hosts compared to 1.111 Å. We conclude that reasonable placement of the link atoms along the C–C bond does not critically affect protonation and deprotonation energies. In their ONIOM implementation combining different levels of QM methods, Morokuma and co-workers also found that the placement of link atoms does not have a pronounced effect on C–H bond dissociation energy.⁴⁹ The placement of link atoms, however, have a more pronounced effect on geometry and vibrational frequency.⁴⁹

V. CONCLUSION

Problems with the most common QM/MM partition methods, demonstrated here for rotational barriers and proton exchanges of small model systems, have been traced to the representation of critical electrostatic interactions between the MM and QM regions. In particular, exclusion of the MM host atom charge from the QM calculation, required in the single link atom method because of the proximity of the quantum link atom, adversely influences these energies. Errors in proton affinities can be 30 kcal/mol or larger for uncorrected SLA treatments.

With the single link atoms, improved proton exchange energies are obtained, when the charges of all MM atoms in the neutral group to which the MMH belongs are excluded from the QM calculation (EXGR option in CHARMM). The improvement is traced to a more balanced electrostatic field for the QM calculation, than when the charge of the MM host atom only is neglected. The EXGR option may, though, omit important electrostatic interactions. This is illustrated for isopropyl alcohol or t-butyl alcohol with multiple interface regions.

The DLA and DGMM methods presented here circumvent the problems with electrostatic interactions. Double link atoms with Gaussian blur of MM charges yield improved rotational barriers for butane and improved proton exchange

energies for model molecules. This is due to improved modeling of the overall charge and dipole. The optimum value of Gaussian blur width, σ , is between 1 and 2 Å. Most chemical bond lengths are also between 1 and 2 Å, and this probably explains why blurring with a σ within this range mimics the actual charge distribution. Protonation energies seem to be robust at σ of 1 Å and deprotonation energies at σ of 1.7 Å. As a consequence of the blurring, it is possible to include the charge of the MM host atom, which otherwise has to be neglected, in the QM calculation. It should be emphasized that the Gaussian blur need not be limited to link atoms, but can be used with other forms of bond termination like frozen orbital or pseudopotential methods.

Our implementation does not require any new parametrization. There is no need to exclude charges selectively. The interface code takes advantage of the QM methods already present in GAMESS-UK and the parameters present in CHARMM. The existing basis sets and methods can be used without further manipulations or parametrizations, and an extensive range of *ab initio* methods can be used. Of course, improvements can be achieved with the fine tuning of QM/MM interface parameters, such as special QM/MM van der Waals coefficients, adjustment of stretching coefficients between QMH and MMH atoms, and with small charge shifts between the MML and the MMH atoms. The results presented here do not include any such fine tuning and demonstrate the quality of the baseline from which minor improvements can be achieved.

The delocalized Gaussian MM charges and double link atom methods avoid problems with the representation of critical electrostatic interactions that plague other QM/MM methods. We expect these methods to be even more critical in an enzyme environment, where there are many more severed bonds and more electrostatic interactions between the QM and MM regions than in the model systems studied here. We advocate using these methods together.

¹R. Baer and M. Head-Gordon, J. Chem. Phys. **109**, 10159 (1998).

²Y. H. Shao, C. A. White, and M. Head-Gordon, J. Chem. Phys. **114**, 6572 (2001).

³M. S. Lee, P. E. Maslen, and M. Head-Gordon, J. Chem. Phys. **112**, 3592 (2000).

⁴C. L. Brooks, B. M. Pettitt, and M. Karplus, Adv. Chem. Phys. **71**, (1988).

⁵M. J. Field, J. Comput. Chem. **23**, 48 (2002).

⁶R. Rajamani and J. L. Gao, J. Comput. Chem. **23**, 96 (2002).

⁷X. Lopez, D. M. York, A. Dejaegere, and M. Karplus, Int. J. Quantum Chem. **86**, 10 (2002).

⁸N. Diaz, D. Suarez, T. L. Sordo, and K. M. Merz, J. Phys. Chem. B **105**, 11302 (2001).

⁹Q. Cui and M. Karplus, J. Am. Chem. Soc. **123**, 2284 (2001).

¹⁰K. P. Eurenium and B. R. Brooks, unpublished results. For example, butane may be partitioned between C1–C2, and between C3–C4 atoms to produce a QM CH₂CH₂ and two MM CH₃ fragments. Without using link atoms, one ends up with either a singlet or a triplet CH₂CH₂. Both the singlet and the triplet states predict the wrong rotational barrier profile. For triplet CH₂CH₂, the *gauche* structure is predicted to be more stable than the *trans* structure by more than 100 kcal/mol at UHF/STO-3G level.

¹¹U. C. Singh and P. A. Kollman, J. Comput. Chem. **7**, 718 (1986).

¹²M. J. Field, P. A. Bash, and M. Karplus, J. Comput. Chem. **11**, 700 (1990).

¹³K. P. Eurenium, D. C. Chatfield, B. R. Brooks, and M. Hodoscek, Int. J. Quantum Chem. **60**, 1189 (1996).

¹⁴HYPERCHEM, Hypercube Inc., Waterloo, ON N2L 3X2, Canada.

¹⁵Y. Zhang, T. Lee, and W. Yang, J. Chem. Phys. **110**, 46 (1999).

¹⁶I. Antes and W. Thiel, J. Phys. Chem. A **103**, 9290 (1999).

- ¹⁷D. M. Philipp and R. A. Friesner, *J. Comput. Chem.* **20**, 1468 (1999).
- ¹⁸J. Gao, P. Amara, C. Alhambra, and M. J. Field, *J. Phys. Chem. A* **102**, 4714 (1998).
- ¹⁹W. Thiel, Program MNDO94, University of Zürich, 1995.
- ²⁰M. J. S. Dewar, E. G. Zoebisch, E. F. Healy, and J. J. P. Stewart, *J. Am. Chem. Soc.* **107**, 3902 (1985).
- ²¹W. J. Hehre, L. Radom, P. v. R. Schleyer, and J. A. Pople, *Ab Initio Molecular Orbital Theory* (Wiley, New York, 1986).
- ²²R. G. Parr, and W. Yang, *Density-Functional Theory of Atoms and Molecules* (Oxford University, New York, 1989).
- ²³*Modern Density Functional Theory: A Tool for Chemistry*, edited by J. M. Seminario and P. Politzer (Elsevier, Amsterdam, 1995).
- ²⁴B. R. Brooks, R. E. Bruccoleri, B. D. Olafson, D. J. States, S. Swaminathan, and M. Karplus, *J. Comput. Chem.* **4**, 187 (1983).
- ²⁵W. D. Cornell, P. Ciepak, C. I. Bayly *et al.*, *J. Am. Chem. Soc.* **117**, 5179 (1995).
- ²⁶P. Sherwood, in *Modern Methods and Algorithms of Quantum Chemistry*, Proceedings, 2nd ed., edited by J. Grotendorst **3**, 285 (2000).
- ²⁷F. Maseras and K. Morokuma, *J. Comput. Chem.* **16**, 1170 (1995).
- ²⁸S. Humbel, S. Sieber, and K. Morokuma, *J. Chem. Phys.* **16**, 1959 (1996).
- ²⁹M. Svensson, S. Humbel, R. D. J. Froese, T. Matsubara, S. Sieber, and K. Morokuma, *J. Phys. Chem.* **100**, 19357 (1996).
- ³⁰D. Rinaldi, J.-L. Rivail, and N. Rguini, *J. Comput. Chem.* **13**, 675 (1992).
- ³¹G. Ferenczy, J.-L. Rivail, P. R. Surjan, and G. Naray-Szabo, *J. Comput. Chem.* **13**, 830 (1992).
- ³²V. Thery, D. Rinaldi, and J.-L. Rivail, *J. Comput. Chem.* **15**, 269 (1994).
- ³³N. Reuter, A. Dejaegere, B. Maigret, and M. Karplus, *J. Phys. Chem. A* **104**, 1720 (2000).
- ³⁴R. B. Murphy, D. M. Philipp, and R. A. Friesner, *J. Comput. Chem.* **21**, 1442 (2000).
- ³⁵D. Bakowies and W. Thiel, *J. Phys. Chem.* **100**, 10580 (1996).
- ³⁶B. Waszkowycz, I. H. Hillier, N. Gensmantel, and D. W. Payling, *J. Chem. Soc., Perkin Trans. 2* **1991**, 225.
- ³⁷B. Waszkowycz, I. H. Hillier, N. Gensmantel, D. W. Payling, *J. Chem. Soc., Perkin Trans. 2* **1991**, 1819.
- ³⁸B. Waszkowycz, I. H. Hillier, N. Gensmantel, D. W. Payling, *J. Chem. Soc., Perkin Trans. 2* **1991**, 2025.
- ³⁹R. J. Hall, S. A. Hindle, N. A. Burton, and I. H. Hillier, *J. Comput. Chem.* **21**, 1433 (2000).
- ⁴⁰I. Logunov and K. Schulten, Proceedings for the Ecole de Physique des Houches on Quantum Mechanical Simulation Methods for Studying Biological Systems, 1995.
- ⁴¹M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids* (Oxford University Press, Oxford, 1996).
- ⁴²GAMNESS-UK is a package of *ab initio* programs written by M. F. Guest, J. H. van Lenthe, J. Kendrick *et al.*
- ⁴³M. W. Schmidt, K. K. Baldrige, J. A. Boatz *et al.*, *J. Comput. Chem.* **14**, 1347 (1993).
- ⁴⁴A. D. MacKerell, Jr., D. Bashford, M. Bellott, *J. Phys. Chem. B* **102**, 3586 (1998).
- ⁴⁵C in a CH₂ group has a charge of -0.18 units, C in a CH₃ group has a charge of -0.27 units. H has a charge of 0.09 units.
- ⁴⁶A. D. Becke, *J. Chem. Phys.* **98**, 5648 (1993).
- ⁴⁷C. Lee, Y. Yang, and R. G. Parr, *Phys. Rev. B* **37**, 785 (1988).
- ⁴⁸LoBoS is the acronym for Lots of Boxes on shelves. See <http://www.lobos.nih.gov>
- ⁴⁹S. Dapprich, I. Komáromi, K. Suzie Byun, K. Morokuma, and M. J. Frisch, *J. Mol. Struct.: THEOCHEM* **461-462**, 1 (1999).